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Core concepts of designing and how we might name them

Per Galle[1]

Abstract

Fundamental questions of what it is *to design,* and doing so *creatively*, are revisited and analyzed in depth. The aim is to develop well-founded insights and core concepts of value for the disciplines of design. Using these insights and concepts to clarify current terminological confusion is a related aim. Drawing on ideas from analytic philosophy, a novel method of incremental conceptual analysis is used to propose new answers to the questions of what it is to design (creatively), in the form of *concept definitions*. Combined with terminology theory, these definitions are used to propose a systematic design-specific *terminology* as a tool for design discourse where precision is important. By way of conclusion, debate about these findings is encouraged, and possible applications in design teaching and practice are briefly indicated. Research applications in ethics, epistemology, and methodology of design are also suggested.

Keywords

creative design, design education, design research, design terminology, philosophy of design

Introduction: Aims and approach

The launch of *Designing*, the new journal of the Design Research Society, is a perfect occasion to revisit the most fundamental questions pertaining to the disciplines of design research, practice, and education: What is it *to design,* and to do so *creatively?*

Trying to settle these questions once and for all would be foolhardy (see, for example, Blackler et al., 2021; Kendall, 2014; Whiting, 2021), indeed downright impossible. But what *is* possible is to make a constructive contribution to a debate about the design concept for the potential benefit of our disciplines. Hence, for this article,

> **Aim no. 1** is to develop a non-trivial concept of design of potential value for research, practice, and education, by proposing a thoroughly justified answer to the above fundamental questions.

Concepts and terminology are two sides of the same coin, because terminology as understood here is a system of concepts defined for a domain (in our case, that of the design disciplines), and terms for naming those concepts.

In situations where clear thinking and precise communication are essential, a domain-specific terminology offering well-chosen concepts and intuitively understandable terms for them would be useful as a reference guide. Unfortunately however, the word "design" can be used in several confusing ways, without it being clear what concept (if any) it refers to. So, once I have presented a solution to the problem of attaining Aim no. 1, I will attend to the other side of the coin, using the resulting concepts of *what it is to design (creatively)* as a seed from which to grow a small systematic terminology that involves a handful of other core concepts as well. Hence,

> **Aim no. 2** is to propose a terminology with definitions of core concepts for the design domain, and intuitively understandable terms for naming them.

I shall refer to that terminology as the *Core Vocabulary System for Designing* (*CVSD*). I have no wish (nor authority) to propose CVSD as an official technical standard. I offer it as a proof-of-concept demonstration of how in the design disciplines we might clarify our conceptual and linguistic resources more thoroughly than we have so far. Yet, I hope some readers will find CVSD useful as a reference guide for serious writing and discourse about design, and I believe that some of the core concepts defined in CVSD will suggest avenues of further research, for example, about epistemology and methodology of design.

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In the subsection below I substantiate and discuss the claim about confusing uses of "design." Background theory and method considerations needed to pursue the dual aim are presented in the next two subsections, from vantage points of terminology theory and of philosophy, respectively. In the subsection "Relation to existing research ..." I position this article in a context of work by others, focusing on what contributions are new regarding method and results. The fifth and last subsection outlines the rest of the article.

Motivation: Why the need for a terminology?

Consider John Heskett's example of a "seemingly nonsensical sentence" (2002/2005: 3):

> Design is to design a design to produce a design.

Nonsensical though it certainly seems, Heskett rightly points out that in this sentence "every use of the word ['design'] is grammatically correct." For, as he goes on to explain in the passage below (bullet points and emphases added):

- The first is a noun indicating a general concept of a field [of *professions and their work*] as a whole, as in: "Design is important to the national economy."
- The second is a verb, indicating *action* or process: "She is commissioned to design a new kitchen blender."
- The third is also a noun, meaning a concept or *proposal*: "The design was presented to the client for approval."
- The final use is again a noun, indicating a finished product [*artifact*] of some kind, the concept made actual: "The new VW Beetle revives a classic design."

In his own words Heskett's point is that "'[d]esign' has so many levels of meaning that it is itself a source of confusion" (2002/2005: 3). Confusion obviously arises from the extreme *ambiguity* of the term "design" so glaringly exposed by his example.[1] In practical communication, of course, no one would seriously utter a sentence like Heskett's. And the *context* in which a particular term is used can often eliminate its ambiguity. But the fact remains that the term "design" is used to name several different concepts (Anne Lise Kjær, 2025, personal communication), which may jeopardize clarity of design thinking and discourse.

So, a normative assumption motivating this article is that we should minimize this risk and refuse to tolerate any concomitant conceptual untidiness. That is why I couple Aim no. 1, about *concept development*, with Aim no. 2 about suggesting *terminological reform*.

Terminology: Domain-specific concepts defined, and terms for naming them

Concepts are tools for thinking. To benefit fully from thinking, not least professionally, we must communicate our thoughts to others, sharing well-chosen concepts and terminology.

Regarding terminology, I shall proceed on the following assumptions and conventions. In this subsection, use of the italicized terms essentially follows ISO standard 1087 (ISO, 2019), from which the expression quoted below was also taken.

For the purposes of professional communication of thoughts within a specific *domain* in situations where precision is important, a *special language* is needed; that is, a *natural language* (such as English) augmented with "specific linguistic means of expression." Prominent among these is a domain-specific *terminology*, which for the purposes of this article we can understand as a system of *concepts,* each of which is assigned one or more *terms*, that is, names of the concept. A concept is represented in the terminology by a *definition*, listed together with the term(s).

Philosophy: Explications, conceptual engineering, and conceptual analysis

Terminologists often *describe* current domain practice of using and naming concepts, drawing on domain experts (Madsen, 1999: Ch. 7), and then try to reduce ambiguity by recommending new terms and concise concept definitions (Madsen, 1999: 76, 82). I share the ambition to reduce linguistic ambiguity, but also want to *improve* our understanding of the very nature of design by developing well-justified in-depth concept definitions not necessarily reflecting current practice.

Therefore, definitions in CVSD will be *normative*, of the type known among philosophers as *explications* (Carnap, 1950: 3–8; Gupta, 2015: Section 1.5; Swartz, 2010: Section 10). An explication seeks to strike a balance between respecting ordinary parlance and intuition on the one hand, and improving the tidiness and precision of (domain-specific) language on the other, so as better to serve scientific or professional purposes.

My mission in this article could be described as an instance of *conceptual engineering*, an early example of which was proposed for design computing (Galle, 1996). The term "conceptual engineering" also turns up in contemporary analytic philosophy. For example, Cappelen (2018) describes several approaches to conceptual engineering and finds it difficult to capture the concept succinctly. But Thomasson aptly characterizes, or perhaps defines, conceptual engineering as "the work of (re-)designing concepts to better serve certain functions" (2017: 364). However, in the context of the present article it might be better to change this into "the work of (re-)*defining* concepts to better serve certain functions."

This being as it may, developing explications is one way of doing what is more commonly known as *conceptual analysis* in (analytic) philosophy. Philosophers do not *agree* on what conceptual analysis is or how it should be done (any more than design researchers agree on what design is or how it should be done); but debating, and improving each other's explications, or suggesting new ones, is certainly what *some*

philosophers do when they perform conceptual analysis. The huge literature on the concept of *knowledge* is a rich source of examples. For a thorough review of the status and disputed nature of conceptual analysis, see (Margolis and Laurence, 2023: Section 5), and of the equally disputed nature of concepts, see their Sections 1–4.

As noted above, I believe in the usefulness of explications. To develop them for CVSD, I will not rely exclusively on abstract intuition (as in "armchair philosophy") but often let arguments be guided by examples of, or related to, design practice. These will be treated only briefly, though; complete case studies are beyond the scope of this article.

Relation to existing research: Novelty, influence from others, and limitations

Definitions of design have been debated for many years. So what *new contributions* does this article offer, and what *influential publications or traditions* helped it underway, and limited its scope? Properly surveying the state of the art of defining design would require an article of its own; suffice it to consider some key examples of existing research to answer these questions.

New contributions obviously comprise the outcomes explicitly aimed at: a carefully developed design concept covering an important class of phenomena, and the derived core vocabulary system proposed for "terminological reform" as a tool for precise communication about design.

For the concept development cum conceptual analysis I use an *incremental* method, which I believe is novel, too. Its point of departure is a concise and prima facie plausible definition or characterization of the concept. That initial statement is subjected to critical scrutiny (in the manner of analytic philosophy), which results in an improved statement with a justification. The new statement, in turn, undergoes further scrutiny and improvement, resulting in yet another justified statement. If successful, the method generates a steady progression of incremental improvements until a satisfactory definition emerges. Each step is relatively simple (hence manageable for reader and author alike), even when the final definition turns out to be complex.

The specific choice of initial statement is not essential. For this article, presumably a good dictionary definition of "design" would have worked. However, I chose Simon's dictum, "Everyone designs who devises courses of action aiming at changing existing situations into preferred ones" (Simon, 1969/1996: 114), because it clearly is "a concise and prima facie plausible definition or characterization of the concept" of design; and because it is widely known in the design community, so many readers will be familiar with it. In making this choice, I furthermore (literally) follow Friedman's recommendation of Simon's dictum as "a good *starting point*" for theory construction in design research (2003: 508, emphasis added).

As for *influential publications or traditions,* analytic philosophers have published several respectable proposals for definitions. The earliest example I am aware of is by philosopher and architect *Greg Bamford*:

> Someone, S, designs or formulates a design for some logically possible thing, A (or type of thing, T_A) at some time, t, just when, (1) S imagines or describes A (or T_A) at t; (2) S supposes in (1) that A (or some token[2] of T_A) would be such as to at least partially satisfy some set of requirements, R, for A (or T_A) under some set of conditions, C; (3) The partial satisfaction of R that S supposes in (2) is a problem for which … (4) … the solution candidate [that] S imagines or describes in (1) is novel for S at t. (1990: 234, ellipses in original.)

As Bamford explains, his four conditions are as follows: "Description, Function, Problem-Solving, and Novelty," each of which does a job of its own. For example, (4) is needed to exclude a case of "[c]hoosing a garden shed from a manufacturer's catalogue."

Like most philosophical work, his definition can be subject to debate: some might find logically possible things ontologically disquieting, for example. But this is not the place for nit-picking philosophical arguments. Bamford's definition set a high standard of precision and depth in the philosophy of design. It works by explicitly stating necessary and sufficient conditions for something to fall under the defined concept; and it uses a modified natural language influenced by formal logic. As you will see later, I adopted these features, albeit in a different form.

Much like Bamford's definition, the comprehensive "reconstruction of product designing" by Houkes and Vermaas (2010: 34–37) is a model of philosophical conceptual analysis of design. Again, there are pros and cons, but another feature I adopted and owe to them is the appeal to designers' *belief formation*.

Like other philosophers I will cite later, Bamford, Houkes, and Vermaas all draw on the analytic tradition of philosophy, and they work within the long-standing tradition of understanding design as basically goal-oriented. I share these predilections and assumptions, acknowledging they limit the scope of my results—as indeed any other predilections and assumptions would do.

The time-honored view of design as goal-oriented has many contemporary manifestations. One is the Modernist idea of functionalism (Galle, 2016: 327–329; Parsons, 2016: 54–68). Other examples are Gero's "Function-Behaviour-Structure Model of Designing" (1990), or Roozenburg and Eekel's contention that "[t]he kernel of [product-design] is reasoning back from statements on functions to statements on the form of the product" (1995: 55). However, other theorists complain that this allegedly dominating element of rationality excludes phenomena they want to call "design." For example, "a practice of self-design and self-curating […] within the context of social media" (Leddy, 2025: 11); and users of designed objects engaging with them in ways not anticipated by the designer (Gal, 2025). While such phenomena are interesting, I find that for the *term* "design(ing)" to be useful, we should reserve it for naming a narrower concept.

Overview of the article

In the next section, "What is it to design?," the incremental method of conceptual analysis is applied to Simon's dictum,

and as a result we get a justified and much more satisfactory *key definition*, of what it is *to design*.

In the section "What is to design creatively?" I take the analysis one step further, extending the key definition so as to explore what it is to design *creatively*. This attains *Aim no. 1*.

The section "Core vocabulary ..." is where I unfold CVSD, the Core Vocabulary System for Designing, deriving additional related definitions (short and simple) from the key definition and its extension. This attains *Aim no. 2*.

Finally, in the concluding section, I look back at the main results, suggesting debate about and applications of them, and indicating avenues of further research.

What is it to design?

In pursuit of Aim no. 1, let's get down to considering the question "*What is it to design?*" ("Design" here is an *intransitive* verb: grammatically, one that does not take a direct object.)

Since at this stage we don't have CVSD at our disposal yet, I will rely on careful use of context to disambiguate any terms that might otherwise cause confusion. However, time and again I use some reasonably self-explanatory term that you will later find integrated in CVSD.

Taking off from Simon's dictum

We begin by rendering Simon's dictum (quoted under "Relation to existing research" earlier) in a style of conceptual analysis, using necessary and sufficient conditions:

> **(1)** An agent **designs** if and only if the agent devises courses of action aiming at changing existing situations into preferred ones.[3]

It would be unfair to criticize (1) as a *definition* of a concept, for Simon did not claim that his dictum was. The passage of his book where it occurs was a criticism of post-WWII professional training at universities and "[s]chools of engineering, as well as schools of architecture, business, education, law, and medicine." Such training, Simon contends, focused too much on natural sciences and mathematics, while failing to acknowledge the role that designing (in the very broad sense of devising "courses of actions …") is playing, or rather *should* play, in all such professional training (Simon, 1969/1996: 111).

Simon's dictum had such a concise wording and persuasive elegance that it might be tempting to accept (1) without further ado as a key definition to build CVSD on. But if taken at face value (1) would not only include typical cases such as an urban designer planning how to change a slum into an attractive residential area, but also phenomena such as "the neighbor's cat planning when and from where to jump at the mouse she has spotted in my garden, or me contemplating an impulse to kick off my shoes under the conference table because my feet are getting hot" (Galle, 2011: 92). Including mouse hunting and shoe kicking as instances of designing would be violating one of Carnap's criteria for a good explication: that its redefined concept be "similar to the explicandum," that is, the original pre-scientific (everyday) concept (1950: 3–8).

A few pages further on in Simon's book we find a passage that could be added as a supplement to (1). "The natural sciences are concerned with *how things are*," he says and (after some remarks on logic that need not concern us), he adds:

> **(2)** Design, on the other hand, is concerned with *how things ought to be*, with devising artifacts to attain goals (Simon, 1969/1996: 114, emphases added).

The "devising artifacts"-clause narrows down the scope of application of "to design" to a more reasonable range, excluding cases like mouse hunting and shoe kicking. But Simon seems to be talking about two different acts: "devising artifacts" in (2) suggests the *actual making* of artifacts, as opposed to the *merely preparatory* act in (1) of devising *a course of action* for improving a situation (presumably by making artifacts later).

Bringing out this distinction is important, because those acts can, but need not, be performed by the same agent: The making of a ceramic teapot may have been prepared *and* carried out by the ceramicist, whereas the architect who designed a building rarely built it, too.

Furthermore, the *kind of preparation* needs clarification. An important part of preparation is to convince oneself (and usually others) that there is *a particular way for an artifact to be*, so that if made that way it will "attain goals."

Based on these considerations about *agent(s)* and about the *nature of preparation*, we combine the gist of (1) and (2), thus arriving at (3):

> **(3)** An agent *D* **designs** if and only if *D* forms the belief that there is a particular way for an artifact to be, such that some [possibly different] agent is able to make an artifact that way.

Simon uses "goal" and "purpose" synonymously (1969/1996: 5). So if we construe "to attain a goal" as "to serve a *purpose*," then goal attainment is implicitly conveyed by the term "artifact" in (3). For, following Hilpinen, I conceive of an artifact as "an object that has been intentionally made or produced *for a certain purpose*" (1999/2011, emphasis added)—albeit with the caveat that artifacts are not necessarily material but may also include intangible entities such as services, interfaces, organizations, and software (Buchanan, 1998, 2001, 2004; Krippendorff, 2007).

I added the concept of *belief formation* in (3), because the definition should not presuppose that any artifact is actually made. Designing only requires that, according to the designer's belief, "some agent" *can* make it. In ordinary parlance we do not hesitate to use the verb "to design" about situations where (quite elaborate) beliefs are formed about "ways for an artifact to be," but where no such artifact is ever made. For example, in 1953 Jørn Utzon, later renowned for his Sydney Opera House, made a very unconventional entry for an architectural competition for a prominent harbor-front restaurant near The Little Mermaid in Copenhagen—but

failed to win the commission (Weston, 2008: 48–55). Yet, there is no denying that what he did on that occasion was to *design*.

Thus, we stick to Simon's contention that *to design* is to prepare for the making of an artifact, and to Hilpinen's idea that an artifact is something intentionally made for some purpose—but adding the further caveat that an artifact may be made for *more than one purpose*.[4]

For example, consider a road bridge crossing a railway line. The engineering team that designed it will have seen to it that once the bridge was there, it would be able to carry the road traffic safely and efficiently across the tracks, leaving the required space for trains to pass underneath. That would be the (main) purpose of the bridge. Additional purposes might be to attract good taxpayers to the local community by improving its infrastructure, or to reduce CO_2-emission from cars of local residents of a village, by providing quick access to the shopping center at the other side of the railway.

However, goals may include more than the purpose(s) ultimately to be served by an artifact. Presumably the team would also have taken into account that the bridge should be comparatively easy and economic to construct and to maintain; and might have made further provisions to ensure that after demolition of the bridge its materials be recyclable. But these considerations are not about *purposes* of the bridge. One does not build a bridge *in order to* maintain it, or *in order to* recycle its materials after demolition. Neither, presumably, are aesthetic considerations about how to make the bridge elegant, harmonically proportioned, or impressive. Yet, elegance, harmony, and impressiveness may well have been among the *goals* that determined what the bridge would be like (the "way for it to be"). So, making sure an artifact would serve its purpose(s) is obviously a goal that had to be attained by its designer; but there may have been many other goals as well.

We should look for a reformulation of the definition that preserves the appeal to artifacts (as we saw, designing is always to prepare for the making of an artifact), but without the bias towards serving purposes. Simply re-introducing Simon's term "goals" might lead us to this:

> **(4)** An agent *D designs* if and only if *D* forms the belief that there is a particular way for an artifact to be, such that (a) some [possibly different] agent is able to make an artifact that way, and (b) any artifact made that way attains ipso facto [i.e., as a result of being made that way] some goals.

In (4), it is clearly an *artifact* that is meant to attain the goals, whereas Simon's original formulations equivocated between attributing the goals to the artifact or to the designer. In ordinary parlance, however, the verbal phrase "to attain goals" would suggest some kind of conscious action. So, in general, speaking of *artifacts* made in a particular way so as to "attain certain goals" seems to strain the language, at the expense of clarity (for a similar observation and critique of Simon, see Kroes, 2002: 295–296).

A better way to express what artifacts should achieve (or rather, what *we* should achieve by means of them) would be to say that they should have certain *valued dispositions*; that is, dispositions (capabilities) that, taken together constitute a *raison d' être*. For example, some valued dispositions that would have motivated the making of an artifact, might be:

- its disposition to be *used* in a certain way (a bridge being used for carrying traffic across an obstacle; a book for providing entertainment or learning, and perhaps for expressing the social status of its owner as well; a teapot being used for making and pouring tea; a festival for promoting a cause and attracting attention to a city; a typeface for composing highly legible text);
- its disposition to be *handled* in certain ways other than use (a particular printer's disposition to be efficiently assembled, packaged, shipped, maintained, and ultimately disposed of in an environmentally responsible manner; or the disposition of a coat to resist wear and tear); and
- its disposition to be *appreciated* in ways that distinguish it favorably from other artifacts of the same or a similar kind (the bridge being unusually elegant; the book offering more suspense or deeper learning than many other books in the genre; the teapot being easier to hold by one hand than other teapots of the same size, the printer being comparatively easy to use, and being more compact than others of similar printing capacity).

Thus, expressing condition (b) of (4) in terms of valued artifact-dispositions (rather than goals) will lead to a clarification.[5]

However, just as goals and valued artifact-dispositions are two closely related concepts, so are such dispositions and *functions*, and the latter has generated a substantial literature.[6] So why complicate matters by introducing "valued artifact-dispositions" as a new term in the definition, rather than using the more well-established term "functions?" There are three reasons for this.

- Among function theorists there is considerable disagreement on what the nature of functions is (Crilly, 2010). For example, is a function of an artifact analogous to the function of the heart in an animal? Are artifact functions the result of selection just as biological functions emerge from survival of the fittest? Some philosophers claim they are. Bamford (2021) rejects this selection theory, and defends a more traditional view of functions as *intended use*. I tend to agree, but will not engage in those debates. This is one reason for preferring valued artifact-dispositions to functions.
- The term "valued artifact-dispositions" is more self-explanatory than "function," partly because of the controversies about functions.
- Despite the diversity of definitions of *function* in the literature, they generally seem biased towards artifact *use*; and, as we saw, such dispositions as being easily maintainable or suitable for recycling are relevant, but go beyond use. Therefore, the concept of *function* seems too narrowly applicable.

Also condition (a) in (4) needs clarification. It requires that the designer forms a belief that "some agent is able to make an artifact" in the way under consideration. But, strictly speaking, there is no need for such an agent to be available *at the time of designing*. For example, it took the Swedish manufacturer some 3 years to develop, to Utzon's satisfaction, the matt and the glossy ceramic tiles for the weather-proof surface of the shells of the Sydney Opera; and it took the contractor another year of experimentation to devise a casting method for embedding the tiles into the top surface of the mass-produced exterior cladding panels (Utzon, 1965/2008). Without these elaborate preparations (and many more) no contractor would have been able to construct the building.

So, it is sufficient for the designer to believe that some agent is, or *can become*, able and motivated to do the job. To acquire the requisite *ability*, the contractors of the Sydney Opera would have had to be informed about Utzon's intentions, and be equipped with the right materials, panel production facilities, and casting method; and as for being *motivated* to put that ability to use, they would have needed some commercially viable contractual agreement with the client.

Expanding condition (a) accordingly, and changing in (b) the appeal to "goal" attainment as discussed above, yields this definition:

> **(5)** An agent *D* **designs** if and only if *D* forms the belief that there is a particular way for an artifact to be, such that (a) some [possibly different] agent is, or can become, able and motivated to make an artifact that way, and (b) any artifact made that way has ipso facto some artifact-dispositions valued by *D* [e.g., dispositions to be used, handled, or appreciated].

Note that in the two clauses of (5), we ascribe *ability* to an agent, and *disposition* to an artifact. I think of both abilities of agents and dispositions of artifacts as *dispositional properties* (a.k.a. capabilities, powers, capacities, potentials, tendencies). What exactly dispositions are, and how they differ from other properties—if indeed they do—are complex questions very much under debate in the philosophy of science, where dispositions figure prominently (for a good introduction, see for example, Schrenk, 2017: Ch. 2–3).[7]

This is not the place to go into those questions. Suffice it to mention that dispositional properties are conventionally characterized by a *triggering condition*, and a *manifestation*. A common textbook example is the disposition of a lump of sugar to dissolve in a bowl of water. The lump actually being dissolved in the water is what manifests the disposition. What triggers it is more complex. Obviously, the sugar must be immersed in the water to be dissolved by it. This is a necessary condition for the sugar to dissolve—but not a sufficient condition. Lots of other conditions must be satisfied for this to happen: the water must not be saturated by sugar already dissolved in it; it must not have frozen to solid ice; it must not have been spilled on the lawn by a clumsy dog knocking over the bowl, and so forth. In short, a disposition only manifests itself under the "right" conditions—but what exactly these are may be difficult to specify.

For the sake of accuracy, I will continue to use "ability" (or ascription of ability: "is able") about the disposition of an agent *to act intentionally* in a certain way. In all other cases, I stick with the use of "(artifact-) disposition". For example, when talking about the disposition of an artifact to be used, handled, or appreciated by an agent, and more generally, *to interact with somebody or something*.

Such artifact-interaction may *allow* an agent some action (a door allows you to pass through a wall), allow some event to take place or state-of-affairs to be maintained (a foundation allows a building to be erected and supported), or it may *prevent* an action or some other event (a banister prevents a person from stumbling into a stairwell; a safety-valve prevents a boiler from exploding).

An artifact may even have a certain degree of autonomy that could be considered a kind of non-human agency: a grandfather clock may chime every hour, or an autopilot may steer an airplane automatically. Still, even when working in these ways, an artifact interacts with human agents, who are *using* it. For a thorough discussion, in a moral setting, of the pros and cons of ascribing agency to artifacts, see (Parsons, 2016: Section 7.4).

In clause (b) of our evolving definition, we will maintain (here and henceforth) the phrase "ipso facto" to indicate that the valued artifact-dispositions are those that an artifact has in virtue of the way it is made. This is to rule out the many irrelevant "valued dispositions" that an artifact would have irrespectively of the way it was made. For a chair, say, the disposition to remain where it is unless moved by some force; the disposition to remain identical to itself even if it rains; the disposition not to grow poisonous tentacles if kicked by a child, and so forth. All of these we might "value" if we thought of them, but having them should not be enough for an artifact to satisfy the condition of clause (b).

Furthermore, in clause (b), we are not talking about *arbitrary* valued dispositions. Designers *care* about the dispositions of artifacts that may be made as a fruit of their work—although the designers may not always be aware of all such dispositions from the outset. Some valued dispositions may have been specified in advance, for example, by authorities issuing rules and regulations (e.g., building codes), or by a client stating needs and wishes in a design brief (Bamford, 2021: Section 3). Others may have occurred to the designer by sudden insights along the way: this is known as *co-evolution* of design problem and design solution (where the "problem" need not be about avoiding an unpleasant or harmful situation, but may just as well be about attaining "valued artifact-dispositions").

Design researchers have studied such co-evolution extensively (see, for example, Archer, 1979; Cross, 2006: 80; Dorst, 2019; Dorst and Cross, 2001; Harfield, 2007; Schön, 1983: 100). Co-evolution involving further aspects of

designing has also been studied (Crilly, 2021a, 2021b; Crilly and Firth, 2019; Halstrøm and Galle, 2014).

Excluding anomalies and introducing artifact-proposals

Our definition now characterizes quite well such acts whose agents, in virtue of performing them, are *obviously* designing, according to ordinary usage of the verb "to design." But the same could be said of Simon's original dictum, and its rendering in (1). Yet, we saw that it was too permissive, contrary to Carnap's requirement that a good explication should respect common parlance as far as possible.

What about (5)? It was developed to exclude mouse hunting and shoe kicking, but there might be other undesirable cases—even anomalies—that it fails to exclude.

Suppose that a designer—*John Doe*—had laboriously formed a belief that there is a particular way for (say) a corkscrew to be, such that conditions (a) and (b) were satisfied. John would have been designing, according to (5). But suppose furthermore that he invites me for dinner, opens a bottle of wine, and in so doing tells me that, as it happens, the elegant corkscrew he is using is the fruit of this recent work of his. Then (provided I understand what he is saying and showing to me), simply by listening to John, *I* too, would have formed the same or a very similar belief: that there is a way for a corkscrew to be, such that (a) and (b) are the case (assuming I value the same dispositions as my friend John). So what I would have done on that occasion would *also* have been to design. Or so (5) tells us.

This is unacceptably at odds with ordinary usage of the verb "to design." The definition is still too permissive. To fix it, we need an additional condition to restrict the kind of belief formations that should count as designing. Adding a condition (c) solves the corkscrew problem:

> **(6)** An agent *D* **designs** if and only if *D* forms the belief that there is a particular way for an artifact to be, such that (a) some [possibly different] agent is, or can become, able and motivated to make an artifact that way; (b) any artifact made that way has ipso facto some artifact-dispositions valued by *D* [e.g., dispositions to be used, handled, or appreciated]; and (c) no artifact has been made that way before.

But we are not home safe yet. Recall Jørn Utzon's unsuccessful 1953 competition entry for a restaurant in Copenhagen (Weston, 2008: 48–55). It is a matter of fact that by reading about it in Weston's book, I formed the belief that indeed there was a particular way for a restaurant to be (viz. as proposed by Utzon), such that a competent and willing contractor could have been instructed and enabled to build a restaurant that way; and such that the resulting artifact would have had "certain dispositions valued by the agent"—that is, by *me*. This would mean that conditions (a) and (b) were satisfied. However, since Utzon's proposal was never realized, condition (c) would have been satisfied too. So according to (6), I could claim to have been "designing" merely by reading about Utzon's restaurant proposal. This, of course, would be just as absurd as thinking I had been designing when John Doe showed me his corkscrew.[8]

The malfunctioning of (6) in cases such as these could be fixed somewhat ad-hoc by changing condition (c) to "no act of making an artifact in that way has been proposed or initiated before." However, nothing much is gained by this, for with or without the ad-hoc modification of (c) the definition still admits as "designing" any act whereby (to quote Boris Hennig, who pointed this out to me) "some agent A forms the belief that *another* agent B knows of a particular way for an artefact to be" such that conditions (a), (b), and (c) are satisfied (2014, personal communication).[9] This would amount to agent A forming "the belief that there is a particular way for an artifact to be"—a belief that satisfies the three conditions—even though A may have no clue as to the "particular way for an artifact to be" that B is supposed to know.

The upshot of Hennig's objection is that one can "design," according to (6), merely by forming the belief that there is a way for an artifact to be—without knowing *what* way, or even what *kind* of artifact—such that the three conditions are satisfied by one's belief.

To see this it is not even necessary to imagine a second agent B. Suppose that a child says, "When I grow up, I'll invent a cool new gadget, to be made by the millions in a big high-tech factory, and people will queue up to buy it, and once they have it, they'll play with it all the time!" Provided this sincerely reports an outcome of belief formation (naïvely self-assured or prophetic as the case may be), would not the child by that very act have been "designing," according to the definition?

It would seem, then, that the salient point regarding novelty is not whether or not an artifact has actually been made in a certain way, but whether or not someone has *thought about* making it that way. However, stating condition (c) in terms of other peoples' thoughts would render our definition impractical, for it might be difficult or impossible for an agent to decide whether or not to believe that anyone else has been thinking—perhaps entirely in private—about the same way for an artifact to be, as the agent him- or herself.

But it is quite reasonable to expect that a competent designer—typically professional—is up to date on recent *public* developments in the practice of the profession, thus being well informed about *artifact-proposals* that have become publically known (e.g., via published competition entries, exhibitions, trade magazines, publications on design history, etc. as in the Utzon restaurant case). These considerations lead us to (7).

> **(7)** An agent *D* **designs** if and only if *D* is making an artifact-proposal, forming the belief that (a) some [possibly different] agent is, or can become, able and motivated to make an artifact according to the proposal; (b) any artifact made according to the proposal has ipso facto some dispositions valued by *D* [e.g., dispositions to be used, handled, or appreciated]; and (c) no such proposal has been made before.

The two new features are the notion of the agent making an *artifact-proposal*, and the reformulation of the

three conditions in terms of such a proposal.[10] In (7) I furthermore use the present participle "is making" rather than "makes" to suggest that to design is not only about the completion of a proposal, but usually involves a considerable amount of hard intellectual work, many failed attempts, etc. (I do not preclude the possibility that an artifact-proposal may emerge in a flash of imagination, or accidentally, for example, as a by-product of playing or doodling.)

Artifact-proposals are often communicated to artifact makers through sketches, working drawings, scale models, CAD files, building information models, specification documents, etc. These examples suggest design of material artifacts, but as noted earlier not all artifacts are material. A composer and an orchestra can be thought of as designer and maker, respectively: the composer's score for a symphony is an artifact-proposal, and the artifacts made according to it by the orchestra would be the performances of the symphony. The performances might have dispositions anticipated by the composer: say, for causing an emotional response or appreciation in an audience.

We should also allow for cases where designer and artifact maker are one and the same. In such cases the artifact-proposal may be something considerably more "private," intended for the designer-cum-maker's own use in artifact-making.

For example, I might communicate to myself the "artifact-proposal" for a new flowerbed in my garden, by marking an area of the ground with loose stones or twigs to suggest an approximate shape and location. It is even conceivable that the artifact-proposal remains entirely implicit, expressed outwardly only through the actual making of the artifact for which it is a proposal. In the flowerbed case, once I had *imagined* the shape and location of the flowerbed, I might seize my garden mattock and proceed directly to the hard work of clearing the area, preparing the soil for planting, etc.

But also more refined artifacts may be made by their designer, without an explicit proposal, and without a sharp distinction between artifact-proposing and artifact-making: A vase, for example, of sophisticated ornamentation and delicate shape, may have been "proposed" by a glass artist confident of her craft, by working with a variety of materials and tools in her workshop. Very likely, such work will proceed by an iterative process of imagination, experiments, and reflections on their outcomes. Saying that an artist "designs" in virtue of doing this kind of work is arguably consistent with (7), although it may constitute a *borderline case*.

Mundane designing

One phenomenon that (7) should also allow for (environmentally problematic though it is) is the *mundane designing* so common for commercial reasons of "product differentiation" and sales maximization. Such work sustains a steady stream of "new" consumer products that differ only marginally from products already on the market: for example, a lightweight jacket lined with red satin instead of last year's black, and fitted with an additional button for good measure; a toothbrush boasting the most high-tech flexible or streamlined handle; an item advertised as a "classically styled large garden bench"; or the latest "lightweight and nimble trail shoe."

Artifacts of this kind are the fruits of proposal-making and belief formation that count as designing according to (7), even though the proposals in question barely manage to satisfy the novelty condition (c). But they *do* manage, because designers making them could claim, correctly, that (as far as they are aware) no "proposal has been made before" *precisely* like these. Condition (c) is satisfied in such cases, too.

What the above examples would suggest is not that our definition is too permissive in regard to mundane designing, but merely the insight that *novelty of proposals need not be impressive* in designing—although novelty cannot be totally absent either, as demonstrated by the anomalous examples of being shown a corkscrew, or reading about Utzon's restaurant proposal.

Delusions

Let us assume that in the corkscrew case I had been in some unhealthy mental condition of extreme envy that caused me, first, to repress my knowledge of John Doe's justified claim to fame for his corkscrew, and then to make an after-the-fact "proposal" for such a corkscrew myself, under the delusion that *I*, rather than John, had originally made that proposal. Would I not then have been "designing" after all, even by the standards of (7)? And furthermore, would I not have been "designing" in the same twisted sense of the word, if, reading about Utzon's restaurant proposal I had managed to deceive myself into believing that *I* was the originator of the proposal—or if by some serious personality disorder I had come to believe that Utzon and I were one and the same person?

The general question raised by these examples is can someone "design" in the sense of (7) by being under a delusion with respect to the novelty condition (c)? As the examples show, an affirmative answer can be defended, which would suggest that the definition admits of certain pathological acts that one might hesitate to describe as "designing."

Definition (7) also admits of pathological cases where the agent is under a delusion about conditions (a) or (b), or even about what it takes to make an artifact-proposal: For example, in bed with flu and a mounting fever, John Doe may think that some random doodle of his will enable his uncle to build a self-replicating robot (a); or he may produce an elaborate annotated diagram (unlike a proper proposal), firmly believing that by following its instructions we can mass-produce affordable flying carpets that will render motor cars redundant (b).

So we have to concede that the concept of designing as so far defined includes certain acts erratically performed due to

(say) psychiatric conditions, drug abuse, brain injuries, or whatever may impair the capacity for making sound judgments. Simple insincerity, carelessness, or laziness might have the same effect.

If you find this concession disquieting, you may wish to interpret the definition in its current form (and all subsequent forms) as tacitly presupposing that the belief it specifies is *rational*, at least to the extent of precluding delusions. Or, in a similar vein, you can read the rest of this article under the assumption that designers (and artifact makers) are *competent* in what they do; that is, mentally sound, sincere, careful and diligent, so as not to succumb to delusions or other conditions causing inadequate artifact-proposals (and artifacts).

We are, after all, trying to define what it is *to design*, not what it is *to design well*. This latter aim we can leave to design methodologists and design educators.

Feasibility, promise, and novelty

When someone designs according to (7) he or she does two things: Makes an artifact-proposal; and forms a belief that the proposal satisfies conditions (a), (b), and (c). We have already been discussing (c) as the "novelty condition." The other two can be named the "feasibility condition" and the "promise condition," respectively. And, without changing the effect of the conditions, we can bring this out more clearly by explicitly stating what it is for an artifact-proposal to be feasible, promising, and novel:

(8) An agent *D* **designs** if and only if *D* is making an artifact-proposal, forming the belief that it (a) is *feasible*, (b) is *promising* some artifact-dispositions valued by *D*, and (c) is *novel.*

(a) An artifact-proposal is **feasible** if and only if some [possibly different] agent is, or can become, able and motivated to make an artifact according to the proposal.
(b) An artifact-proposal is **promising** some artifact-dispositions if and only if any artifact made according to the proposal has ipso facto those dispositions [e.g., dispositions to be used, handled, or appreciated].
(c) An artifact-proposal is **novel** if and only if it differs from any previously made artifact-proposal.

The three sub-definitions in (8) provide additional and useful resources for talking about artifact-proposals as "feasible," "promising," and "novel." Furthermore, the (modest) complexity of (7) becomes more manageable by breaking it down into the short main definition, and three simple sub-definitions of (8).

However, upon closer inspection and further analysis the sub-definitions may not be so simple as they seem. Particularly in (a), more information is crammed into a few words than does justice to the insights on feasibility already gained.

Recall that when discussing (4), we considered in some detail what it took for the contractors to acquire the *ability* to construct the roof shells of the Sydney Opera; namely, "the right materials, panel production facilities, and casting method." Such items are *means & methods of production*, that is, the resources to which an agent must have access in order to make an artifact according to a given proposal.

At the time Utzon made his proposal for the shells in their final form, based on a spherical geometry, the proposal was not yet what I would call *ripe*. That is to say, although Utzon knew what kind of shells he wanted, the necessary agent(s) and means & methods of production were not yet available for making an artifact according to his proposal. For a proposal to be *ripe*, an agent must be available (contractor and/or manufacturer) who has access to the relevant means & methods of production, and is able and motivated to use them for making an artifact according to the proposal at hand.

But even though Utzon's proposal for spherical shells was not ripe from the outset, it was already what I have called "feasible," for Utzon and his co-workers were able and motivated to *make* the proposal ripe. They did so, of course, through negotiations and co-operation with manufacturers and contractors, whom they motivated to develop the requisite means & methods of production.[11]

To unfold the complexity of feasibility illustrated by the Sydney Opera case, and describe it in case-independent terms, we introduce an auxiliary condition (a') and reformulate (a) in terms of (a'), as shown below.

The three properties that an artifact-proposal has when it satisfies clauses (a), (b), and (c) we can talk of as its *feasibility*, its *promise*, and its *novelty*, respectively.

(9) An agent *D* **designs** if and only if *D* is making an artifact-proposal, forming the belief that it (a) is *feasible*, (b) is *promising* some artifact-dispositions valued by *D*, and (c) is *novel.*

(a) An artifact-proposal is **feasible** if and only if: either it is *ripe*, or else some agent is available who is able and motivated to make it *ripe*.
(a') An artifact-proposal is **ripe** if and only if some agent and some means & methods of production are available, such that the agent is able and motivated to make an artifact according to the artifact-proposal using those means & methods.
(b) An artifact-proposal is **promising** some artifact-dispositions if and only if necessarily, any artifact made according to the proposal has ipso facto those dispositions [e.g., dispositions to be used, handled, or appreciated].
(c) An artifact-proposal is **novel** if and only if it differs from any previously made artifact-proposal.

Definition (9) is equivalent to (8), only more explicit—except that in clause (b) I have added "necessarily" (a so-called modal operator). It modifies the sentence at the right-hand side of "if and only if." This is to fix a logical problem with the promise condition that would otherwise occur. Let

me explain it briefly, and non-technically, by means of an example.

Imagine clause (b) *without* "necessarily," and suppose that during an act of design John Doe had made a *feasible* artifact-proposal for a perpetual motion device, forming the belief that any artifact made according to his proposal will have the disposition to produce useful work without consumption of energy. (Something we know today is physically impossible.[12]) Then, as long as no artifact has been made according to John's proposal, it would indeed be true what he believes! But, alas, only for purely technical reasons of logic. This is an example of what is known in logic as "vacuous truth." As soon as a device were made according to John's proposal, what he believed would treacherously become false. So, although his belief at the time of designing would have been true, it was not reliable!

A concept of *promise* in artifact-proposals that malfunctions because of vacuous truth is clearly useless. But adding the modal operator "necessarily" in clause (b) fixes the problem. That is because the operator requires the sentence that follows it to be true *no matter what happens* in the future.[13] Or, if you like, to be true *in all possible worlds*.[14]

Problems of vacuous truth do not arise for *feasibility* and *novelty* of artifact-proposals, because they are exclusively about what already exists at the time of designing.

Let me close this section by pointing out that any plausible definition of designing must allow for *failure*; for designers may be mistaken with respect to feasibility, promise, or novelty of any artifact-proposal they are making:

- A proposal for a railway tunnel connecting London and Canberra along a straight line through the Earth would be *novel* and *promising* (saving time and energy as compared to surface traveling), but obviously fails to be *feasible*.
- A proposal for a perpetual motion device may be both *feasible* and *novel*, but will fail with regard to *promise* (no knives will be sharpened, nor other useful mechanical work done, without consumption of energy). The proposal for the Tacoma Narrows Bridge, which collapsed in 1940, is another case in point.
- A proposal for an elegant logo may be both *feasible* and *promising*, but it might fail to be (sufficiently) *novel*, so the designer could end up sued for plagiarism.

Yet, in such cases it would be quite appropriate to speak of designing. As it should, (2.9) accommodates the possibility of failure by its appeal to *belief formation*—in virtue of the simple fact that beliefs can be wrong.

What is it to design creatively?

As we have already seen, there is an entire *spectrum* of design(ing), ranging from the very mundane (streamlined tool brush, classical garden bench, etc.), over cases involving a considerable degree of original thinking (the unusually elegant corkscrew) or a high degree (Utzon's unrealized restaurant project), to the cases of extraordinary originality (the Sydney Opera House). These cases are all instances of designing (i.e., the concept *to design*) as defined so far. But what is it *to design creatively?*

According to Boden, the "core notion" of the "many definitions of creativity" that philosophers and psychologists have offered, is "the capacity to generate ideas or artifacts that are both *new* and *positively* valuable" (2005: 477, emphases added). Gaut concurs: "There is a broad consensus that creativity is the capacity to produce things that are *original* and [positively] *valuable*" (2010: 1039). The value condition is indispensable, he contends, because without it, producing "original nonsense" would count as creativity.

Translated into the language of (9), the upshot of these views (which I share) is that for designing to be *creative*, the agent (designer) making an artifact-proposal must form the belief that

- the *artifact-dispositions* promised by the artifact-proposal has *positive value*, and
- the *artifact-proposal* has a *strong* (*high degree of*) *novelty.*

Let us consider these two aspects one at a time.

Positive value of artifact-dispositions

Designers will normally strive to attain artifact-dispositions that they themselves, or their clients, regard as positive. But any designer should keep two observations about values in mind:

- *Values of artifact-dispositions may be agent-relative:* What counts as positively valued artifact-dispositions for the designer or the designer's client may count as negatively valued artifact-dispositions for others. A torture instrument, or a rifle, may have dispositions regarded as positive by designer and client. But obviously they are negative from the point of view of the victims against whom such artifacts are used. A car's disposition to move fast is of great convenience to its owner, but is a nuisance to cyclists, pedestrians, and stray hedgehogs.[15]
- *Values of artifact-dispositions may be mixed:* What a designer or client regards as positive dispositions might come along with other dispositions of negative value. The disposition of a sewing machine to stitch together pieces of cloth is accompanied by dispositions to hurt the hands of its user, to produce dust and noise, etc.

A designer's ability to take these facts into consideration when making an artifact-proposal is a hallmark of professional competence. However, that does not call for modification of definition (9), for as noted earlier, our aim is to clarify the concept of *designing*, not to develop methods for *designing well*.

Strong novelty of artifact-proposal

To define what we mean by "to design creatively," all we need to do is modify definition (9) by strengthening (c), the novelty condition. We simply introduce the concept of *strong novelty* of an artifact-proposal, which is like novelty except for the additional requirement that the proposal differ *considerably* from proposals previously made. As our discussions of mundane designing suggest, there is no sharp boundary between what is mundane designing and what is not; trying to be precise about just *how much* an artifact-proposal must differ from any predecessors for it to be strongly novel, would be futile.

Since designers can be mistaken about the novelty of their artifact-proposals, the kind of creativity we are dealing with is only what Boden (2004: 2, 43 ff.) calls "psychological creativity" (involving novelty-*belief*). It is not what she calls "historical creativity" (involving novelty as a matter of historical fact). But the more thoroughly and successfully a designer tries to achieve novelty when making an artifact-proposal, the closer to historical creativity his or her designing will be.

Definition (10) below is the extended key definition mentioned in the article overview at the end of the introduction. It is identical to (9) except for the four additions in curly brackets. They yield a concept definition of what it is "to design creatively" along the lines suggested above.

(10) An agent *D* **designs** {**creatively**} if and only if *D* is making an artifact-proposal, forming the belief that it (a) is *feasible*, (b) is *promising* some artifact-dispositions valued by *D*, and (c) is {*strongly*} *novel.*

(a) An artifact-proposal is **feasible** if and only if: either it is *ripe*, or else some agent is available who is able and motivated to make it *ripe*.
(a') An artifact-proposal is **ripe** if and only if some agent and some means & methods of production are available, such that the agent is able and motivated to make an artifact according to the artifact-proposal using those means & methods.

(b) An artifact-proposal is **promising** some artifact-dispositions if and only if necessarily, any artifact made according to the proposal has ipso facto those dispositions [e.g., dispositions to be used, handled, or appreciated].

(c) An artifact-proposal is {**strongly**} **novel** if and only if it differs {considerably} from any previously made artifact-proposal.

In a study by Gomes et al. (2022) they propose three metrics for *evaluating* design creativity, in terms of numerical scores regarding what they call *feasibility*, *utility*, and *novelty* of design solutions. Here, "design solutions" corresponds to my term "artifact-proposals," and the three metrics bear a certain conceptual similarity to my triple properties in (10) of *feasibility*, *promise*, and *novelty* of an artifact-proposal. Despite considerable differences in aim, method, and definitions the conceptual parallels between their metrics and my conception of creative designing are worth noting. The fact that these ideas were conceived and developed independently of one another might support a claim as to the plausibility of both.

Core vocabulary system for designing (CVSD)

Building on key definition (9) and its extension (10) we now unfold our proof-of-concept terminology, CVSD, as per Aim no. 2. Each entry consists of reference number, one or two ("core") terms, a concept definition, and one or two examples to illustrate usage of the term(s).

(11) Term: **design** [intransitive verb].

Concept definition: (9).

Example: "We all *design* to some extent every day. We assemble our place of work, our home, and even the way we look" (Lawson, 2004: 7; italics added, here and in subsequent examples).

(12) Term: **design** [transitive verb. *Preferred use:* when direct object refers to actual artifact.]

Concept definition: An agent **designed** an artifact if and only if the agent *designed* (11) and the artifact was made according to the resulting artifact-proposal

Example: "Sir Joseph Paxton *designed* the Crystal Palace …" (Walford, 1878).

Example: We are *designing* the new town hall to be built soon. [Discouraged.][16]

(13) Term: **designer** [noun, countable].

Concept definition: An agent is a **designer** if and only if the agent *designs* (11).

Example: "Each *designer* creates a bespoke piece that offers a contemporary take on a classic film" (British Fashion Council, 2012).

(14) Term: **act of design** [nominal phrase, countable].

Term: **design** [noun, countable].

Concept definition: An act is an **act of design** (a **design**) if and only if the act is performed by an agent who thereby *designs* (11).

Example: To perform *an act of design* creatively, you must free yourself of tradition.

Example: "Inspired by Joseph Paxton's *design* of Crystal Palace, Brunel hired the same contractors …" (Design Museum, 2012).

(15) Term: **design** [noun, countable. *Usage discouraged*].

Concept definition: An entity is a **design** if and only if the entity is an artifact-proposal made by an *act-of-design* (14).

Example: "*Designs* [i.e., (artifact-)proposals] on display at degree shows or in portfolios, often showed detailed research …" (Thomas et al., 2011: 4).[17]

(16) Term: **designing** [noun, uncountable – to be distinguished (by context) from "designing," present participle of the verb "to design" (11).]

Term: **design** [noun, uncountable].

Concept definition: An act is an instance of **designing** (of **design**) if and only if the act is an *act of design* (14).

Example: "*Design* [a.k.a. *designing*] is important to the national economy" (Heskett, 2002/2005: 3).

Example: "*Design* [a.k.a. *designing*] for me is about finding the most appropriate way to do something …" (Business Link, 2012).

The following entries, involving creativity, are derived from the extended key definition (10).

(17) Term: **to design creatively** [intransitive verbal phrase].

Concept definition: (10).

Example: "We *design creatively*, build precisely and execute beautifully" (SmartProd, 2011).

(18) Term: **creative designer** [nominal phrase, countable].

Concept definition: An agent is a **creative designer** if and only if the agent *designs creatively* (17).

Example: "… Gerald Benney was a highly *creative designer* keen on strong and simple designs [i.e., (artifact-) proposals], …" (Lyon, n.d).

(19) Term: **act of creative design** [nominal phrase, countable].

Term: **creative design** [noun, countable].

Concept definition: An act is an **act of creative design** (is a **creative design**) if and only if the act is performed by an agent who thereby *designs creatively* (17).

Example: To perform *an act of creative design*, you must free yourself of tradition.

Example: "It is a fact well known that the *creative design* of a city creates an environment conducive to …" (Creative Cape Town, 2012).

(20) Term: **creative designing** [nominal phrase, uncountable].

Term: **creative design** [nominal phrase, uncountable].

Concept definition: An act is an instance of **creative designing** (of **creative design**) if and only if the act is an *act of creative design* (19).

Example: "[*Creative designing*] is important to the national economy" (Heskett, 2002/2005: 3).

Example: "*Creative design* is of course about looking good, but it's a lot more than that; …" (TwoSixTwo, 2012).

Conclusion: Results, applications, and avenues of further research

We found non-trivial argument-supported answers to the questions of what it is *to design,* and doing so *creatively.* The answers are open to challenges and may provoke debate. This should be welcomed, as it would offer an opportunity for the design community to reflect on its very *raison d'être* and future development.

But even as they stand, the answers provided in key definition (9) and its extension (10) have some merit. For,

- They enabled CVSD (see the previous section), a systematic terminology that goes some way to dispel conceptual and linguistic confusion, and may serve as guide, for example, for accurate writing.
- They provide a basis for challenging design students (and professionals) to explain why and how their artifact-proposals are feasible, promising, and (strongly) novel.
- They suggest new research:
  - into design *ethics* (see discussion of values and their agent-relativity earlier, under the heading of "Positive value of artifact-dispositions");
  - into design *epistemology*: how do (or should) creative designers manage to form reliable beliefs about the promise of their proposals?
  - into design *methodology*: how do (or should) designers make trade-offs among promised but incompatible artifact-dispositions, or between promise and feasibility?

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Notes

1. Such ambiguity is caused by *polysemy*: the phenomenon of one word form (or term) carrying multiple meanings. Wüster, pioneer of terminology theory, sought to avoid polysemy entirely, to enable unambiguous (technical) communication; more recently terminologists have taken other aspects of communication into account as well (Castellví, 2003; L'Homme, 2024: 221–222).
2. It would seem that "some token" should have been "any token." (Confirmed by Bamford, 2025, personal communication.)
3. I assume "agent" refers to either individuals, or groups or organizations of individuals working together. For convenience of exposition, I shall generally speak of designers (and other agents) as if they were individuals, but on the understanding that they may not be so in practice.

4. A similar point was made by Crilly (2013) who explored how, in general, an artifact may be seen as part of several systems nested within one another, in each of which the artifact has a system-specific function (function being closely related to purpose).
5. The idea of valued dispositions of an artifact (to be used, handled or appreciated) can be considered a variant of Vitruvius' architectural virtues of *firmitas*, *utilitas*, and *venustas* (strength, utility, and beauty, respectively) which have been around since Antiquity (Kruft, 1994: 21, 24 ff.).
6. Concepts of *function* are invoked in theoretical and philosophical literature on design (and biology). The diversity of function concepts is discussed by Crilly (2010). He considers several classifications of functions. One such classification divides them into *technical*, *social*, and *aesthetic* functions, which does not exhaust all possibilities, however: 32. In my discussion above, valued "dispositions *to be used*" covers both technical and social functions, while aesthetic functions would fall under "dispositions to be appreciated." My examples of valued "dispositions to be handled," however, are not functions at all, since they have nothing to do with use. Important to us are dispositions *anticipated by the designer*—not so much what class they belong to.
7. Theorizing about artifacts in terms of their *dispositions* is not new: Houkes et al. (2002) do so, in a discussion of the so-called "functions" of artifacts, drawing on work by Cummins back in 1975. For example, they write, "the wing of an aeroplane is assigned the function 'providing lift' because it has 'providing lift' as a disposition and because this disposition partly explains the plane's disposition to fly" (318–319). As for *human abilities*, Vetter raises some doubt as to whether they should be regarded a kind of dispositional properties, as some philosophers do (2019). But regarding human abilities as dispositional properties is not essential to my argument.
8. I owe this insight to Peter Kroes (2013, personal communication).
9. Hennig's comment was triggered by a definition that was slightly different but in any relevant way identical to (6).
10. The notion of an artifact-proposal was introduced by Galle and Kroes for similar purposes (2014: Section 3.1), and is reminiscent of a "design representation" (Galle, 1999).
11. Relating these events, I draw on Utzon's own description as cited earlier, and on the comprehensive account of the Sydney Opera project published by Mikami (2001), who was deeply involved in developing its geometrical and structural details.
12. For a fascinating and comprehensive account of attempts to make such machines, dating back to a time when they were still taken seriously, see (Dircks, 1861).
13. For more on the philosophical idea of necessity, see, for example, Nolan's concise introduction to modality (2009: 96), or Garson's book-length treatment (2006/2013).
14. See, for example, (Nolan, 2009: 98–100) for a brief non-technical introduction to possible worlds, or Plantinga (1974: 44–45) for an equally brief and accessible presentation of possible worlds as maximal states of affairs.
15. Similar agent-relativity of values has been described, albeit in different terms and in different ways, in the design-theoretical literature on *affordances* (Gibson, 1979/1986). Affordances (which seem related to dispositions) enjoy widespread popularity among members of the design professions, thanks to Norman's classic, *The design of everyday things* (2013). For an interesting discussion of Gibsonian affordances by an analytic philosopher, see (Vetter, 2020).
16. The discouraged usage may be harmless when unlikely to create misunderstandings. But, *strictly* speaking, a non-referring term such as "the new town hall to be built soon" is ontologically problematic. At the time of designing, no real building existed for the term to refer to, just as no real horse exists for "Pegasus" to refer to.
17. This usage of "design" unnecessarily contributes to the confusion illustrated by Heskett's example. I suggest we use "(artifact-)proposal" instead, which is much more self-explanatory.

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